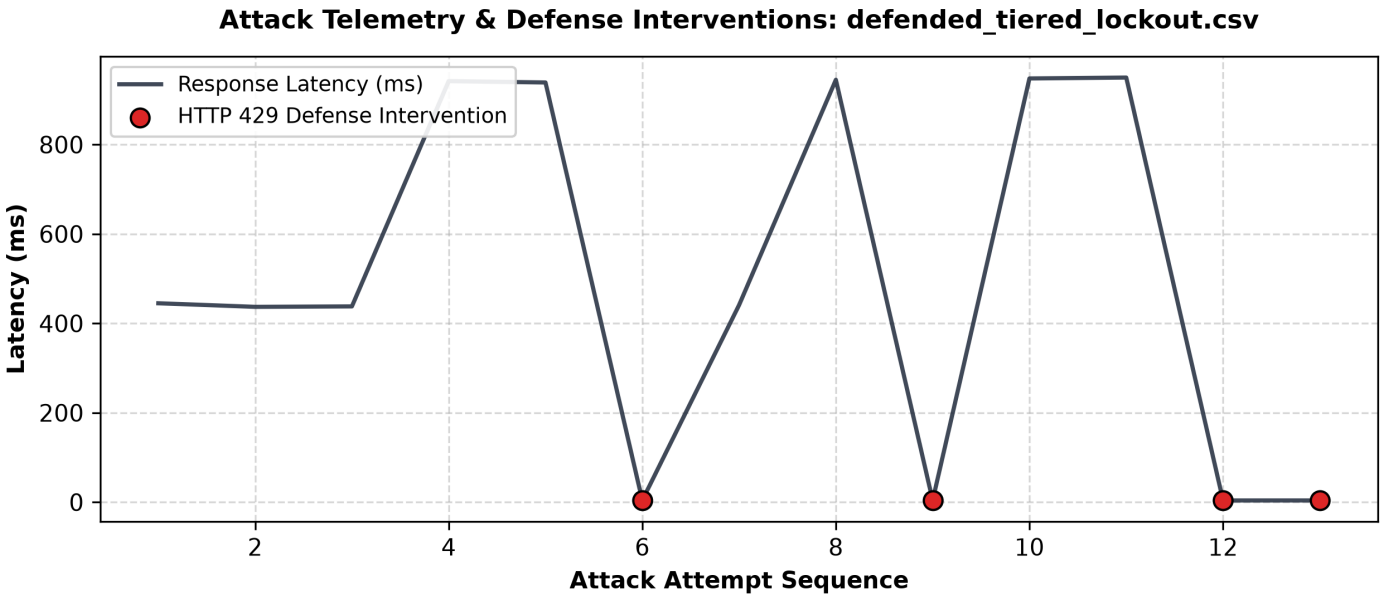
# PROJECT JANUS // FORENSIC AUDIT REPORT

Empirical Attack & Defensive Intervention Assessment • Generated: 2026-10-27 00:19:38

| | | | |
|---|---|---|---|
| **Target Endpoint** | :8080/lab/verify (Android Daemon) | **Audit Outcome** | **DEFENDED (Permanent Lockout Enforced)** |
| **Total Verification Invocations** | 13 | **Defense Interventions (429)** | 4 events |
| **Key Derivation Algorithm** | PBKDF2_HMAC_SHA256 | **Constant-Time Verification** | True |
| **Hardware Enclave Protection** | StrongBox / TEE Hardware Keystore | **Discovered PIN Result** | None (Blocked) |

## EMPIRICAL TIMELINE & LATENCY BEHAVIOR

Real-time observation of device cryptographic verification latency and progressive rate-limiting triggers.




## QUANTITATIVE PERFORMANCE & LATENCY METRICS

| Metric | Observed Value | Security & Operational Significance |
|---|---|---|
| Mean Server Verification Latency | 720.56 ms | Cryptographic computation time inside secure enclave. |
| Median Server Latency (P50) | 939.00 ms | Representative baseline per-request cost. |
| 95th Percentile Latency (P95) | 949.20 ms | Upper-bound latency under synthetic system load. |
| Exhaustive Keyspace Time (10,000 PINs) | PERMANENT LOCKOUT (Infinite) | Time required for an automated adversary to brute-force all 4-digit combinations. |

## FORENSIC EVENT LOG (KEY TRANSITION SEQUENCES)

| Seq | PIN | Status / Event | Latency | Defense Tier / Notes |
|---|---|---|---|---|
| 1 | 0000 | FAILED | 445.0 ms | PBKDF2_HMAC_SHA256 |
| 2 | 0001 | FAILED | 437.0 ms | PBKDF2_HMAC_SHA256 |
| 3 | 0002 | FAILED | 438.0 ms | PBKDF2_HMAC_SHA256 |
| 5 | 0004 | FAILED | 939.0 ms | PBKDF2_HMAC_SHA256 |
| 6 | 0005 | 429 RATE LIMIT | 4.1 ms | TEMPORARY_LOCKOUT_TIER_2 |
| 7 | 0005 | FAILED | 441.0 ms | PBKDF2_HMAC_SHA256 |
| 8 | 0006 | FAILED | 945.0 ms | PBKDF2_HMAC_SHA256 |
| 9 | 0007 | 429 RATE LIMIT | 4.0 ms | EXTENDED_LOCKOUT_TIER_3 |
| 10 | 0007 | FAILED | 948.0 ms | PBKDF2_HMAC_SHA256 |
| 11 | 0008 | FAILED | 950.0 ms | PBKDF2_HMAC_SHA256 |

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***Attestation:*** *This document certifies the empirical security results captured by the Project Janus Automated Test Harness. Cryptographic operations are verified against Android Keystore hardware-backed implementations.*